AG News: TF–IDF and BERT

A paired evaluation audit

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Question and scope

- How much does BERT improve over TF–IDF Logistic Regression in this experiment?
- Which predictions change on the same test examples?
- Single-run academic artifact; no state-of-the-art or peer-review claim.
- Paired statistics audit historical predictions; full BERT fine-tuning was not rerun.

Experimental protocol

Split	Examples	Use
Train	108,000	Fit models/features
Development	12,000	Evaluate baselines; select BERT checkpoint
Test	7,600	Final evaluation

Four balanced labels: World, Sports, Business, Sci/Tech. Stratified 90/10 train/development split, seed 42. Inputs include a headline and description.

Models

- Mode baseline: training labels only.
- Multinomial NB and LR: TF–IDF unigrams/bigrams, English stop words, 20,000 features; fixed settings.
- BERT: bert-base-uncased, 128 tokens, batch size 8, two epochs, AdamW, learning rate 2e-5, weight decay 0.01.
- BERT checkpoint selected by development loss.

Historical test scores

Model	Accuracy	Macro-F1
Most frequent	25.00%	10.00%
Multinomial NB	90.24%	90.19%
Logistic Regression	91.80%	91.78%
BERT	94.25%	94.24%

LR and BERT scores recomputed from saved predictions. Other rows are historical summaries. No measured compute-cost comparison is available.

Paired outcomes

Outcome	Count
Both correct	6,862
LR wrong / BERT correct	301
BERT wrong / LR correct	115
Both wrong	322

Net gain: **186 examples**, or **2.45 percentage points**.

Conditional statistical evidence

- Paired bootstrap 95% interval: **1.92–2.97 percentage points**.
- 10,000 paired resamples; seed 42.
- Exact two-sided McNemar p: 2.71×10^{-20}.
- Conditional on fixed saved models and exchangeable test examples.
- Does not measure variation across training seeds or prove a causal mechanism.

Gross corrections are not net gains

- Business/Sci-Tech: 100 of 301 BERT-only corrections.
- Net Business/Sci-Tech error reduction: **40 of 186** (21.5%).
- Net World/Business reduction: 53.
- Net World/Sports reduction: 42.
- Gains are spread across topic pairs; most do not concentrate in Business/Sci-Tech.

Class-level gains

Gold class	LR correct	BERT correct	Gain
World	1,730	1,802	72
Sports	1,864	1,883	19
Business	1,674	1,718	44
Sci/Tech	1,709	1,760	51

Each class has 1,900 examples. World has the largest absolute gain; Sports is close to ceiling.

Length is descriptive, not causal evidence

Partition	Median words
All inputs	37
BERT correct	37
BERT wrong	36
LR wrong / BERT correct	35

Similar medians cannot rule out length effects or establish context understanding. Analyze length bins, truncation and controlled text changes next.

Research software improvements

- Offline audit: metrics, paired statistics, input hashes and descriptive CSVs.
- Strict pairing; stable example IDs for new runs.
- Fresh output directories preserve historical artifacts.
- Save/reload classical pipelines and BERT; JSON headline predictions.
- Offline unit and training/inference integration tests; CI configuration.
- Editable report/slides and document build script.

Current replication check

- Full classical rerun: NB **90.16%**, LR **91.88%** test accuracy.
- Historical values: NB 90.24%, LR 91.80%; test texts/labels match.
- LR predictions differ on 74 examples; original metadata is insufficient to explain why.
- Real pretrained BERT loaded and completed one optimizer step on training data.
- These checks validate execution; they do not reproduce the full historical BERT experiment.

Limitations and next experiments

- Missing historical checkpoint, run configs, hardware and immutable revisions.
- One run per model; no seed-variance estimate.
- No domain shift, calibration, controlled ablation or measured resource benchmark.
- Next: multiple seeds, fixed revisions, documented tuning budgets, new-domain evaluation and causal probes.

Takeaway

The saved predictions support a **2.45-point accuracy gain** for BERT on this test set. The paired audit makes the evidence inspectable. Stronger claims about generalization, cost and mechanism require new experiments.

Dataset: Zhang et al. (2015). BERT: Devlin et al. (2019). Full references, methods and caveats appear in the revised report.